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ETM 3800

Topic 7

The Category Rules and Syntax

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1. Introduction to Category Rules
2. Category rules and syntax
3. Boolean and Proximity Operators
4. Assessing Categorisation Outcomes

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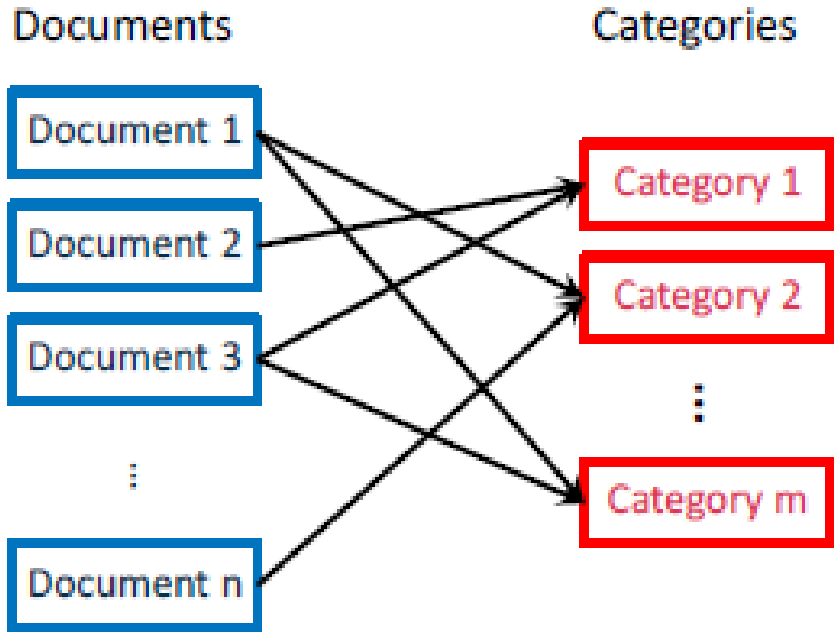
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Introduction to Category Rules

- SAS Visual Text Analytics can categorise the same document into many relevant categories through the Categories node.
- A category identifies a group of documents that share a common characteristic.
- The categories can be generated automatically, the topics added as categories or the variables designated as category variables in the Data Tab.



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Introduction to Category Rules

- Category rules have a single form that uses Boolean and proximity operators.
- Category rules are written as Boolean rules because they resolve to true or false. A value of True results in a match.
- They determine whether a document belongs to a specific category.
- These rules can be difficult to maintain when they get larger.
- However, they are generally faster to run and process than Concepts rules.

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Introduction to Category Rules

- Use Category rules to categorise the documents.
- And use Concepts rules to extract information from documents in your input data.
- Use Category rules in conjunction with Concepts rules when the Category rules are too large to maintain.
- In categories matches on concepts are based on an all matches method, which returns all matches found in the text.
- By contrast, in concepts, matches are based on the best match method.

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Introduction to Category Rules

The best match method determines the match based on the following factors:

- If a span of text matches multiple concepts, the concept with the highest priority value is returned as a match. Custom concepts have a default priority value of 10.
- If a span of text matches multiple concepts, and those concepts have the same priority value, then the concept that produces the longest match is returned.
- If a span of text matches multiple concepts, and those concepts have the same priority value and produce matches of equal lengths, then the first concept that is compiled is returned as the best match.

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Category rules and syntax

(OPERATOR, argument1, argument2, ...)

where arguments can be terms, strings, or nested rules.

General rules for syntax:

- Boolean operator names are specified in uppercase.
- Boolean and proximity operators and their arguments are enclosed in parentheses and separated with commas. The arguments are included in quotation marks (“ ”). Example: (AND, "my_w holiday", "_cap")
- Operators and their arguments can be nested as an argument of another operator. Example: (AND, (OR, "courage", "courageous"), (OR, "brave", "bravery"))

Category rules and syntax

(OPERATOR, argument1, argument2, ...)

where arguments can be terms, strings, or nested rules.

General rules for syntax:

- Reference a category from another category using a special syntax called tmac syntax (_tmac).
- Concept names can be referenced in category rules. If you reference a concept, the concept matches are used to contribute to the true/false match of the category rule. Concept names must be enclosed in brackets ([ ]). For example, to reference the concept gameShows in a category rule, you could write the rule (OR, "[gameShows]")
- Special symbols can be used to modify the rules to include wildcards, case sensitivity, and so on

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Category rules and syntax

- Example, when a Category rule is displayed in text view:

  (AND,"Amazon",(OR,"online retailer","e-commerce"))

- This rule is true when the term Amazon appears with either of the terms online retailer or e-commerce.

Boolean Category rule Examples

- Subcategory: To extract documents containing the term "equation(s)":

  (AND, (OR, "equation", "equations"))

- Subcategory: To extract documents containing the term "measure and relationship":

  (AND, (OR, "measuring", "measures", "measured", "measure"), (OR, "relationship", "relationships"))

- Subcategory: To extract documents containing Business Intelligence or Enterprise Miner ~ (Enterprise Guide, DDE, Drug):

  (AND, (NOT, (OR, "Enterprise Guide", "DDE", "Drug")), (OR, "Business Intelligence", "Enterprise Miner"))

- To specify the previously defined concept rule Business Intelligence (BUSINESS_INTELLIGENCE):

  (OR,"[BUSINESS_INTELLIGENCE]")

Boolean Category rule Examples

- Suppose you have the following category structure under All Categories: NAME / FIRST LAST
- For example, suppose you want to reference the category FIRST to a category called FIRST_NAME.

    Category Name: NAME

    Sub-category Name: FIRST

    (OR,"Terry","Peter","George")

    Sub-category Name: LAST

    (OR,"Woodfield","Christie","Fernandez")

    Category Name: FIRST_NAME

    (OR,_tmac:"@NAME/FIRST")

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Boolean and Proximity Operators

- Refer to SAS Visual Text Analytics 8.5: User's Guide, Boolean and Proximity Operators for Category Rules from pages 133 to 139.

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Assessing Categorisation Outcomes

- **True Positives**: Retrieving the information that was requested.
- **False Positives**: Retrieving information that is unrelated to the request.
- **False Negatives**: Omitting information that is related to the request.
- **True Negatives**: Omitting irrelevant information.

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Assessing Categorisation Outcomes

Notation

ND = Number of documents

NS = Number of selected documents (retrieved)

TP = Number of true positives

FP = Number of false positives

TN = Number of true negatives

FN = Number of false negatives

		Action	
		Retrieved	Omitted
Actual	Contain Info	TP	FN
	Absent Info	FP	TN
	Column Totals	NS	ND-NS

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Assessing Categorisation Outcomes

- Misclassification Rate = Fraction misclassified

  MISC = (FP + FN)/ND

- Precision = Fraction of retrieved documents that are relevant

  Precision = TP/(TP + FP)

- Recall = Fraction of relevant documents that are retrieved

  Recall = TP/(TP + FN)

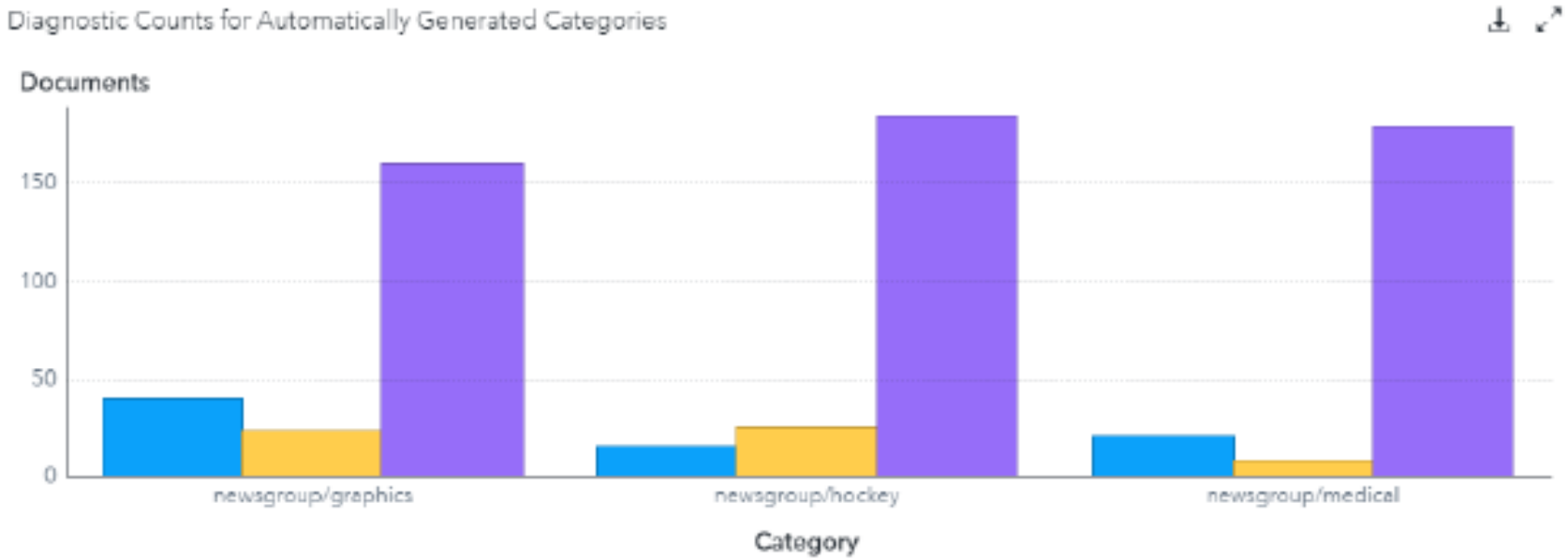
- F1 Statistic (F-Measure) = Harmonic mean of precision and recall

  F1=2/[(1/Precision)+(1/Recall)]=(2*Precision*Recall)/(Precision +Recall)

		Action	
		Retrieved	Omitted
Actual	Contain Info	TP	FN
	Absent Info	FP	TN
	Column Totals	NS	ND-NS

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Assessing Categorisation Outcomes



- The Diagnostic Counts for Automatically Generated Categories chart in The Category node shows document counts for the number of true positives, false positives, and false negatives by category.

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Assessing Categorisation Outcomes



- The **Diagnostic Metrics for Automatically Generated Categories chart** displays the **F-Measure, Precision**, and **Recall** values for each automatically generated category.
- A lower number of false positives results in a higher precision value, and a higher number of false positives results in a smaller precision value.
- The recall value depends upon the number of false negatives present. A lower number of false negatives results in a higher recall value. A higher number of false negatives results in a lower recall value.
- The F-Measure is a reflection of both the recall and precision values. A value between 0 and 1 represents these three measures.

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Lesson Summary

- Categories node produces categories of documents that share a common characteristic.
- The categories can be generated automatically, from topics or designated variables.
- Category rules use Boolean and proximity operators. Therefore, it can be difficult to maintain but run faster.
- Categories and concepts can be referenced in a category rule using tmac syntax and square bracket [ ], respectively.
- The categorisation outcomes can be assessed by misclassification rate, precision, recall and F1 statistic.

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Would you like to have hands-on experience with category rules applications using the Categories node?



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